Lecture 15

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Power series : $\sum_{n=0}^{\infty} a_n (x-c)^n = a_0 + a_1 (x-c) + a_2 (x-c)^2 + \dots$

$a_n \in R$, $c \in R$

$x = x_0$ a fixed real no. $\sum_{n=0}^{\infty} a_n (x_0 - c)^n$

Radius of conv. of power series : R

i) series is abs cgt for $|x-c| < R$ $\underbrace{(c-R, c+R)}_{\text{Region of conv.}}$

ii) div for $|x-c| > R$

$\sum_{n=0}^{\infty} x^n$ $\underline{|x| < 1}$ $1 \to$ Radius & $-1 < x < 1$ Region of conv.

$= \frac{1}{1-x}$ or interval or domain

How to find Radius of conv. $\sum a_n (x-c)^n$

$c = 0$

$\sum a_n x^n$

$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} |a_n|^{1/n} = \frac{1}{R}$

Eg 1) $\sum n! \, x^n$ $c = 0$ $a_n = n!$

$\frac{1}{R} = \lim_{n\to\infty} \left| \frac{(n+1)!}{n!} \right| = \lim_{n\to\infty} n+1 = \infty \Rightarrow R = 0$

$\Rightarrow$ Series div everywhere.

2) $\sum \frac{x^n}{n}$ $c = 0$ $a_n = \frac{1}{n}$

$\frac{1}{R} = \lim_{n\to\infty} \left| \frac{1}{n+1} \cdot \frac{n}{1} \right| = \lim_{n\to\infty} \frac{n}{n+1} = 1 \Rightarrow R = 1$

$|x - 0| < R = 1$

$(-1, 1)$

3) $\sum \frac{x^n}{n!}$ $a_n = \frac{1}{n}$ $c = 0$

$\frac{1}{R} = 0 \Rightarrow R = \infty \Rightarrow$ series cgt everywhere

$\frac{1}{R} = \lim$

$n!$

$$\frac{1}{R} = \lim_{n\to\infty} \frac{1}{n+1} = 0 \Rightarrow R = \infty \Rightarrow \text{series cgt everywhere on } \mathbb{R}$$

4) $\sum \frac{(x-2)^n}{n}$ $\quad c = 2 \quad a_n = \frac{1}{n}$

$$\frac{1}{R} = \lim_{n\to\infty} |a_n|^{1/n} = \lim_{n\to\infty} \frac{1}{n^{1/n}} = 1 \Rightarrow R = 1$$

$|x-2| < 1$

$(1,5) \to [1,3)$

At $x=1$ $\quad \sum \frac{(-1)^n}{n} \to$ cgt

At $x=3$ $\quad \sum \frac{1}{n} \to$ div

Special case of power series: $f(x) = \sum_{n=0}^{\infty} a_n (x-c)^n$

$$= a_0 + a_1(x-c) + a_2(x-c)^2 + a_3(x-c)^3 + \dots$$

$a_n = \frac{f^{(n)}(c)}{n!}$ $\quad f^{(n)}(c) \to n$ th derivative of f at c.

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x-c)^n \quad \text{Taylor Series}$$

$\to c = 0 \to$ Maclaurin series

eg: 1) $f(x) = e^x$ $\quad f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$

Taylor series about $c = 0$

$f'(x) = e^x \quad f''(x) = e^x \quad \dots \quad f^{(n)}(x) = e^x \quad f^{(n)}(0) = 1$

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} (x-0)^n = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \dots$$

2) $f(x) = e^x \quad c = -3/2$

$f'(x) = e^x \quad f''(x) = e^x \dots f^{(n)}(x) = e^x \quad f^{(n)}\left(\frac{3}{2}\right) = e^{3/2}$

$$f(x) = \sum_{n=0}^{\infty} \frac{e^{3/2}}{n!} \left(x + \frac{3}{2}\right)^n = e^x$$

3) $f(x) = x^2 + 3x + 2 \qquad c = 1$

$f'(x) = 2x + 3 \qquad f''(x) = 2 \qquad f'''(x) = 0$

$$f(x) = \underset{\overset{\shortparallel}{f(1)}}{f^{(0)}(1)} + \frac{f^{(1)}(1)}{1!}(x-1) + \frac{f^{(2)}(1)}{2!}(x-1)^2$$

$$= 6 + \frac{5}{1!}(x-1) + \frac{2}{2!}(x-1)^2 = 6 + 5(x-1) + (x-1)^2$$

Taylor polynomial